

SOLUTIONS - Exercise - Week 13: MATH1061/7861

- (1) (5 marks) A system generates user IDs from the integers 1 to 1000. Due to legacy constraints, certain IDs are flagged as follows:

- IDs divisible by 2 are flagged as *internal*,
- IDs divisible by 3 are flagged as *test*,
- IDs divisible by 5 are flagged as *archived*.

A developer defines a *clean ID* to be an ID that is not flagged in any way.

How many clean IDs are there between 1 and 1000?

Let A be the set of IDs that are flagged as *internal*, B be the set of IDs that are flagged as *test* and C be the set of IDs that are flagged as *archived*. We want to find $1000 - |A \cup B \cup C|$.

By the inclusion-exclusion principle,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|.$$

There are $1000/2 = 500$ IDs divisible by 2, so $|A| = 500$. Likewise,

$$|B| = 333$$

$$|C| = 200$$

$$|A \cap B| = 166$$

$$|B \cap C| = 66$$

$$|A \cap C| = 100$$

$$|A \cap B \cap C| = 33$$

so

$$|A \cup B \cup C| = 500 + 333 + 200 - 166 - 66 - 100 + 33 = 734.$$

Therefore there are $1000 - 734 = 266$ clean IDs.

MARKING:

1 mark for identifying the number of internal IDs, the number of test IDs and the number of archived IDs

1 mark for identifying the number of IDs that are both internal and test, and similar sets

2 marks for correctly stating/using the inclusion-exclusion principle

1 mark for a correct final answer

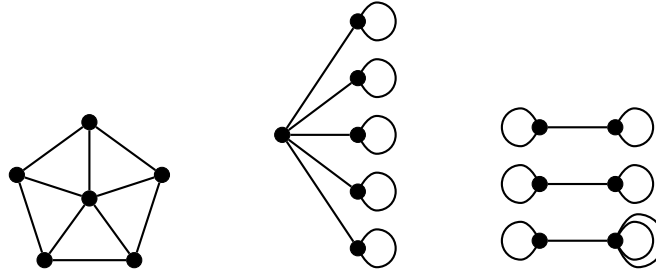
- (2) **(5 marks)** A graph has 10 edges and 6 vertices. Five of the vertices have degree 3.
 (a) What is the degree of the remaining vertex? Explain your reasoning and draw a picture of such a graph.

Using the Handshake theorem, the sum of the degrees is equal to two times the number of edges. If we let the degree of the remaining vertex be x , then

$$2 \cdot 10 = 20 = 3 \cdot 5 + x.$$

Thus, $x = 5$, so the last vertex has degree 5.

There are many ways to draw such a graph, especially since the question does not specify that it must be simple or connected. Here are some examples:



- (b) Does the graph you drew have a circuit?

Their graph will always have a circuit. Students should indicate one on their graph.

- (c) Does the graph you drew have a Eulerian trail?

Their graph will never have an Eulerian trail, as there are more than 2 vertices of odd degree (recall that a graph has a Eulerian trail if and only if it is connected and there is either 0 or 2 vertices of odd degree). Students should explain why one does not exist.

MARKING:

2 marks for determining that the last vertex has degree 5

1 mark for drawing a valid graph

1 mark for identifying a circuit on their graph

1 mark for correctly explaining why their graph does not have an Eulerian trail